Historical Information-Guided Class-Incremental Semantic Segmentation in Remote Sensing Images

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***Abstract*— Despite the extraordinary success of the deep architectures on semantic segmentation for remote sensing (RS) images, they have difficulties in learning new classes from a sequential data stream because of catastrophic forgetting. Continual learning for semantic segmentation (CSS) is an emerging trend for its capability to cope with the above problems effectively. However, old classes from previous steps are collapsed into the background, which further aggravates the challenge of CSS in the RS scene. In this article, we revisit the knowledge distillation (KD) strategy and the characteristics of class-incremental semantic segmentation (CISS) and then present a generalized and effective framework to learn new classes while preserving knowledge of the learned classes. In particular, we propose two novel historical information-guided modules: the feature global perception module and the label reconstruction (LR) module. The former enables the current model to pay more attention to the region related to the old categories identified by the historical information when learning new classes. Meanwhile, the latter retrieves pixels belonging to the learned classes from the background to handle the background shift problem and maintain the high performance of old classes. We have conducted comprehensive experiments on two RS semantic segmentation datasets of Instance Segmentation in Aerial Images Dataset (iSAID) and Gao Fen (GF) challenge semantic segmentation dataset (GCSS). The experimental results outperform the current state-of-the-art methods in most incremental settings, which demonstrates the effectiveness of the proposed framework.**

***Index Terms*— Catastrophic forgetting (CF), continual learning, remote sensing (RS), semantic segmentation.**

I. Introduction

In RECENT years, the acquisition of a large number of high-resolution remote sensing (RS) images brings severe challenges to the existing RS semantic segmentation methods [1], [2]. Most existing methods [3]–[13] are commonly offline training and static deployment. When dealing with new categories or tasks, these approaches either mix all the data and retrain the model or fine-tune the network based on the latest task data. Although retraining can achieve superior performance, the time consumption and memory requirements of the network will become higher as the number of task sequences increases. Fine-tuning (FT) is effective on the current task, but the performance on the previous tasks will drop sharply after learning the new one [14]. Therefore, it is imminent to find a strategy with the capacity to acquire new knowledge while maintaining high performance on previous tasks. Continual learning has gained widespread attention for its ability to solve the above problems.

Continual learning is an algorithm that learns continuously from a sequential data stream where new tasks occur. Based on whether the task identity is provided during the test time, if not, whether it must be inferred, continual learning can be divided into three scenarios: task-incremental learning (TIL), class-incremental learning (CIL), and domain-incremental learning (DIL) [15]. This article explores a more challenging CIL scenario because all tasks in CIL share a single-head output. Single head does not demand the task identity to be known, while multihead does at test time [16]. Compared to the multihead configuration of TIL, single head may bring distraction to model learning. CIL should be able to acquire the knowledge of the new classes while ensuring that the performance of the learned ones does not degrade too much. However, deep learning networks are inherently prone to catastrophic forgetting (CF) [17] due to their architectural design, which constrains the development of continual learning. CF refers to the significant decline in the performance of the model on the previous tasks after learning the new one.

In recent years, many approaches of continual learning have emerged. Following [18], we summarize these methods into three families: replay-based [19]–[26], parameter isolation-based [27]–[33], and regularization-based [16], [34]–[42]. Among them, the regularization-based method has two branches: prior-focused and data-focused. The former applies the distribution on the parameters of the old model as a prior when learning new tasks. The latter makes the new model acquire the hidden knowledge from the old by exploiting knowledge distillation (KD) [43].

The methods of the three families mentioned above have been extensively studied and verified in image classification, particularly data-focused methods. The methods based on data-focused are favored because they do not require additional storage and can retain previous knowledge without using old

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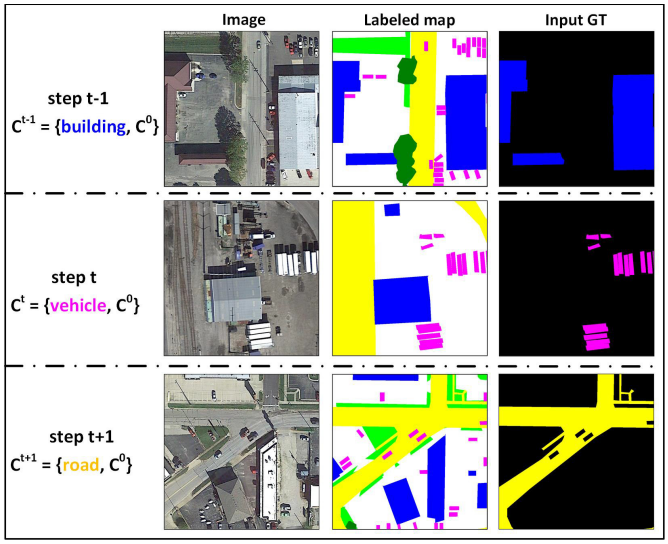


Fig. 1. Semantic shift of background class in ground-truth masks. C^t represents labels provided in the current step, C^0 stands for background, and GT denotes the ground truth. Taking addition one class at a time as an example, the rightmost column is the supervision information input to the network for each incremental step. At step t, background class pixels may contain old classes (e.g., building) and classes to be learned in the future (e.g., road).

data. The previous continual learning methods based on KD mainly contain two modules. One module is employed to preserve the knowledge of the previous model, which is related to the output probability distillation (OPD) loss. The other is applied for learning new knowledge, which corresponds to the commonly utilized cross-entropy (CE) loss in classification. These methods have achieved good performance in the field of image classification. However, they may not be able to solve the CF problem well because of the following two challenges in the class-incremental semantic segmentation (CISS) in RS images.

A. Complex Characteristics of RS Images

The complex characteristics of RS images will increase the difficulty of CISS and exacerbate the CF issue. Compared with natural scene images, RS images are characterized by multi-scale objects, complex scenarios, and foreground–background imbalance. For example, there could be football fields and cars in the same image. Besides, an RS image may contain dozens or even hundreds of objects. In addition, the background in RS images is more complicated, and the same object may appear in different contexts. All these properties and complex RS scenes may exacerbate the CF problem. Therefore, if we can make the model to focus on the learning of foreground-region in the process of CISS, it will help to suppress the influence of complex scenes and alleviate the imbalance between the foreground and the background to a certain extent.

B. Semantic Shift of the Background Class

Semantic shift phenomenon is a peculiar aspect of CF in semantic segmentation [38]. Background class refers to pixels that do not contain any object in the traditional semantic segmentation. However, under the setting of CISS: at each training

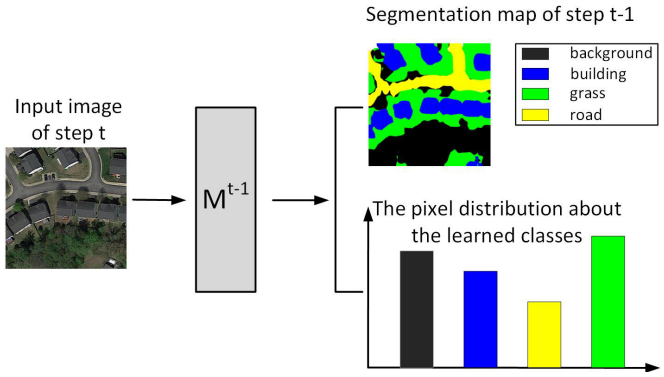


Fig. 2. Illustration of historical information. The input images of step t are processed by M^{t-1} to obtain the predicted segmentation map (class label for each pixel) and pixel distribution (the number of pixels each class label has) of the learned categories, which constitute historical information.

step, only a subset of annotations of all possible classes are accessible, and other classes that do not belong to the current step are classified as background class. Consequently, background class pixels alter as the learning step changes, which is shown in Fig. 1. In other words, the CISS background class contains three types of pixels: the pixels without any object, the pixels that have been learned in previous tasks, and the pixels to be learned in the future. If we can extract learned classes information from the background class without using the old data, it will contribute to alleviating semantic shift and preserving previous knowledge.

Most existing CIL methods ignore the hidden historical information in the background class of the current step. Historical information refers to knowledge about old categories that have been learned in the old model, as shown in Fig. 2. If historical information can be extracted from the background class and put to use, it can help mitigate the CF problem.

To deal with the two challenges, we propose a CIL framework based on historical information, which is extremely important since it is the only source of information about the old classes without annotations. For the first challenge, we construct an HFGP module. To deal with the second challenge, we design an HLR module. The comparison of the continual learning pipeline between previous work and our work can be seen in Fig. 3.

The HFGP module applies historical information to make the feature space having the ability to learn new knowledge while paying more attention to the foreground region of the learned classes. The HLR module obtains support from the historical information and confidence-based pseudo-label strategy. Unlike [38], which incorporates the true background class probability into the computation of CE loss and KD loss, our proposed HLR retrieves the annotation pixels belonging to the old classes from the background class, thereby overcoming the difficulty of background semantic shift to a certain extent.

In order to verify the effectiveness and generalization of our method, we conduct experiments on instance Segmentation in Aerial Images Dataset (iSAID) [44] and GaoFen challenge semantic segmentation (GCSS) dataset [45], [46]. The experimental results illustrate that our framework is the state of the art in most incremental settings compared with

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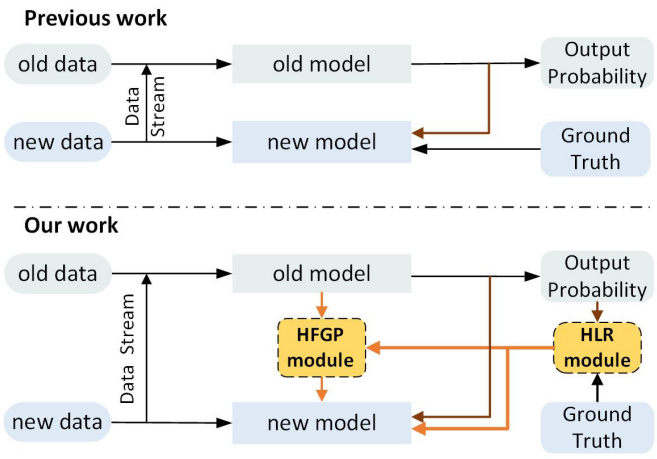


Fig. 3. Comparison of the continual learning pipeline between previous work and our work. In previous works, they only apply the output probability to prevent CF. However, since the lack of preceding classes label information, it may not be adequate to use the output probability to preserve the learned knowledge. In our work, we utilize historical information to reconstruct the GT of the current task. Furthermore, we apply a feature perception module to enhance the focus on the foreground region in the feature. The historical information-guided feature global perception (HFGP) and historical information-guided label reconstruction (HLR) represent the proposed HFGP and HLR, respectively. The brown lines represent output probability flow and the yellow lines denote the flow of relevant information in the framework guided by historical information.

recently continual learning methods. Our contributions can be summarized as follows.

1) This article reveals that the application of historical information at the feature and output levels contributes to the retention of learned knowledge in CIL. Based on the historical information, we propose a CISS framework for RS images.
2) The HFGP module is proposed to equip the model with the ability to focus more on the foreground region belonging to the old model in the feature space, which helps to retain the high performance of learned classes.
3) We combine the historical information with the label reconstruction (LR) strategy and propose the HLR module. This module can retrieve the label pixels belonging to the old classes from the background class, which effectively improves the CISS background shift problem.
4) The experimental results on the iSAID and GCSS datasets illustrate the effectiveness of our framework. As far as we know, this is the first CISS work over several classic continual learning approaches on iSAID and GCSS.

II. Related Works

In this section, we will briefly review some relevant works of semantic segmentation in RS images. After that, we focus on the research status of continual learning, which has been extensively studied for its ability to handle sequential learning tasks.

A. Semantic Segmentation of Remote Sensing Images

With the proposed fully convolutional networks (FCNs) [47], the semantic segmentation methods of RS images have achieved impressive advancements. These methods improve the segmentation accuracy by expanding the receptive field to capture multicontext characteristics [3]–[6] or introducing the attention mechanism [7]–[13]. For instance, work [3] exploits the multicontext information by training a dilated network with a defined patch size. While a hybrid multiple attention network (HMANet) to efficiently and adaptively capture global correlations from multiple aspects is designed in [9]. With the emergence of transformer [48] based merely on the attention mechanism, the segmentation transformer (SETR) [49] employs a pure transformer to encode images as a series of patches, which improves the semantic segmentation performance to a new level. The transformer-based approach has also been applied to the field of RS segmentation. Work [13] first adopted the swin transformer [50] as the backbone to obtain context information for semantic segmentation in RS. While WiCoNet [12] introduces a context transformer to embed context information and selectively project it onto local features extracted with a convolutional neural network (CNN). Although these methods can effectively deal with semantic segmentation, they will suffer CF when encountering sequential learning tasks.

B. Continual Learning in Natural Scene Images

Continual learning and CF have been widely researched in image classification tasks. Existing works can be divided into three families: data replay, parameter isolation, and regularization approaches. Three different branches tackle the problem of CF in different ways.

1) Data Replay Branch: The methods of data replay solve the problem of CF from the data level by keeping a few samples of the previous concepts [19]–[21], [23] or employing the generator to produce the data of the previous tasks [24]–[26]. In incremental classifier and representation learning (iCaRL) [19], the authors simultaneously learn classifiers and a feature representation with a small number of previous training data. In gradient episodic memory (GEM) [20] and A-GEM [21], the authors consider the setting of TIL and solve the problem of overfitting the subset of stored samples via constrained optimization. Pooled outputs distillation network (PODNet) [23] applies a local similarity classifier (LSC) to generate multiple proxy vectors for each class. Unlike the above method which directly uses the data of the previous tasks, deep generative replay (DGR) [24] suggests reconstructing data distribution by training a deep generative model to generate pseudo-data. Hyper-LifelongGAN [26] establishes a generic framework for various types of conditional image generation. However, the methods of replaying the actual data may be unavailable due to privacy issues and a large amount of storage space is required to store data. In addition, the performance of replay-based approaches is highly dependent on the capacity of the memory and the quality of the saved samples. Although using generative adversarial network (GAN) to generate pseudo-data avoids contacting with real data, training GAN continually increases the complexity of the model.

2) Parameter Isolation Branch: Parameter isolation methods are also regarded as network expansion [27]–[33], which deals with CF at the structural level. These approaches expand

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separate network branches for new jobs or reserve a fixed subset of model parameters for each task. For instance, progressive neural networks (PNNs) [27] preserve and fix all the networks for previous tasks and dynamically create a new network when there comes a new task. Random path selection (RPSNet) [32] proposes a novel path selection algorithm that gradually chooses the optimal path for new tasks while encouraging parameter sharing between tasks. As the number of tasks increases, the network volume of expansion methods will significantly grow. PackNet [29] and Piggyback [30] both produce the binary parameter usage mask under a fixed deep neural network (DNN) architecture. In the case of infinite memory, these methods work effectively to overcome the problem of CF. However, the additional parameters of these methods will generate expensive calculations and huge storage requirements as the sequence of learning tasks increases. These methods may not suitable for continuous learning with long sequences. Meanwhile, they cannot be deployed on edge devices where memory and computational resource are severely limited.

3) Regularization Branch: The regularization approaches can be grouped into prior-focused and data-focused. The former methods are concerned with the change of significant parameters that represents knowledge of previous tasks. Elastic weight consolidation (EWC) [34] employs the Fisher information to determine the importance of parameters and imposes severe penalties on changing these parameters when learning new tasks. Synaptic intelligence (SI) [42] conducts an assessment of importance at each synapse during training based on their contribution to change in the learned prediction loss. Riemannian walk (RW) [16] presents a generalization of continual learning algorithms by combining EWC and SI. Incremental moment matching (IMM) [36] approximates the mixture of Gaussian posterior values to the Gaussian distribution of a combined task. However, these methods become infeasible as the number of parameters increases because they assume that the parameters in the neural network are independent and each parameter is required to estimate its importance.

The core of the data-focused methods is to implement KD between the previously learned model and the current training model. Learning without forgetting (LWF) [35] first introduces the KD strategy into the incremental learning paradigm. PODNet [23] and iCaRL [19] integrate this idea with representation learning. Learning without memorizing (LWM) [22] presents an attention distillation loss to punish the change of attention maps. Incremental learning techniques (ILTs) [37] focus on the continual learning for semantic segmentation (CSS) and additionally perform a distillation operation on the intermediate feature layer before the decoding stage. Since ILT [37], more and more researchers have been working on the CSS scenario. A multiscale spatial distillation loss is proposed in [39] to preserve long- and short-range spatial representations at the feature level. In [40], the latent space is regularized to enhance the separation of class-conditional features. In [41], replay images of previously learned classes are retrieved through a web crawler or a generative network, and a self-inpainting strategy is introduced to further label the obtained images. Modeling the background (MiB) [38] considers the problem of background shift characteristic of CSS and proposes a strategy to incorporate the probability of true background class into the calculation of CE loss and KD loss. Differing from MiB, our proposed approach solves the background shift problem by using historical information to reconstruct the old classes' annotations corresponding to the current input images. Then, we use the new labels obtained after reconstruction to calculate the CE loss.

Apart from using the above three families of continual learning alone, there are mixed methods that combine the three types of approaches. In [51], a knowledge base and an active column are combined to form an extensible framework. iCaRL [19] integrates KD with representation learning that employs a small amount of previously seen data. PODNet [23] generates multiple proxy vectors for each class and combines spatial-based KD loss to combat CF, while in [52], a novel KD strategy is proposed and synthetic replay data are used simultaneously.

C. Continual Learning in Remote Sensing Images

Recently, some scholars express their interest in studying continual learning in RS images. Lu *et al.* [33] achieved lightweight continual learning in RS image scene classification by feature transfer. In the CSS, Feng *et al.* [53] paid attention to TIL setting without accessing previous tasks data, while Tasar *et al.* [54] focused on CIL with storing samples of learned classes. In this article, we are concerned about CIL in RS images just like [54]. However, we consider the most realistic but challenging setting where data from previous tasks are not preserved and cannot be used to assist the incremental steps. Because we do not utilize the data of the learned classes, we can only depend on the parameter information retained by the old model for the source of the previous knowledge. Therefore, the effective exploitation of historical information is extremely important in dealing with CF. Different from most methods that only distill the output logits between the old and new model. Inspired by ILT [37], we apply the distillation loss both on the intermediate features and output logits. We integrate historical information into the current model to preserve previous knowledge when learning new categories. In addition, we employ historical information to generate pseudo-labels about previously learned classes to deal with background shift problem in CISS.

III. Methods

In this section, we introduce the historical information-guided framework from four aspects. We first describe the problem formulation of continual learning in Section III-A. After that, the overall network architecture of our model is presented in Section III-B. Finally, in Sections III-C and III-D, we illustrate the proposed two historical information-guided modules.

A. Problem Formulation

CIL is an algorithm that sequentially learns new classes from a continuous stream of data. At any time, the learner

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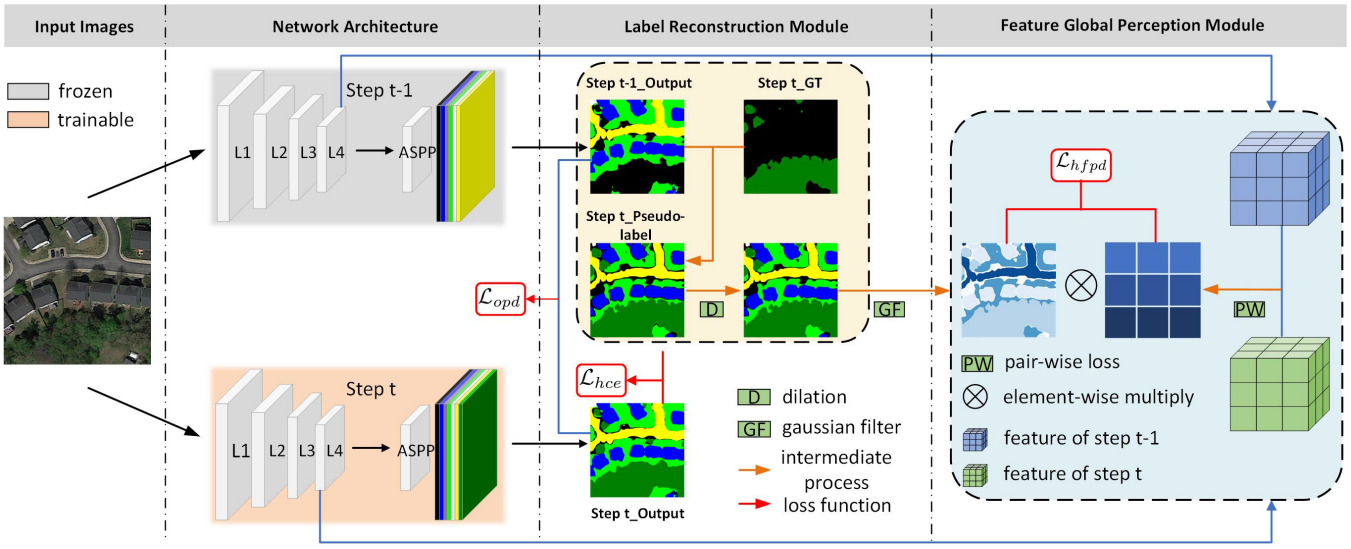


Fig. 4. Overview of the tth incremental step of our historical information-guided framework. During the training step t, input images are fed into M^{t-1} and M^t to acquire the corresponding output logits. By combining $\widehat{P}^{t-1}$ with the GT of step t, we have restored some pixel annotations belonging to the old classes with a high enough confidence. The pseudo-labels obtained by our proposed HLR module and the output of M^t constitute HCE loss, which is applied to learn new classes. The HFPD loss related to the HFGP module in the intermediate feature space along with the OPD loss is used to retain the knowledge of the previous classes. Note that, we do not utilize any exemplars from previously learned categories.

TABLE I
NOTATIONS AND THEIR MEANINGS IN THE PROPOSED METHOD

Symbol	Meaning
$\mathcal{T}$	the number of learning steps
M^t	model learned in step t
Θ^t	all learnable parameters of step t
$f^t(\cdot)$, $h^t(\cdot)$	feature extractor and classifier
$\widehat{P}^t$	predicted segmentation mask of M^t
I^t	the set of input images in step t
W, H	the width and height of input images
C	the classes seen over time
D_{tr}^t	training data of step t
$(x_i^t,\ \hat{y}_i^t)$	the ith input image and GT of step t
$\mathcal{Y}^t$	the set of GT segmentation mask of step t
C^t	the set of classes in step t
c_{bg}	the true background
C^0	the background class of CISS

can perform multiclass classification of all seen classes over time. The training of CIL occurs in multiple stages called learning steps or tasks and each learning step contains one or more categories. Assume that we have a sequence of supervised learning assignments with T steps, expressed by $\mathcal{T} = 1, 2, \ldots, t, \ldots, T$. We denote the model that learned in step t as M^t and the set of all learnable parameters as $\Theta^t = \theta_l^t$, where $l = \{1, 2, \ldots, L\}$ represents different layers of the model. Generally, M^t consists of a feature extractor $f^t(\cdot)$ and a classifier $h^t(\cdot)$. $\widehat{P}^t = \widehat{P}\left(\cdot; \Theta^t\right) \in \mathcal{R}^{W\times H\times C} = h^t(\cdot) \circ f^t(I)$ is defined as the predicted segmentation mask of M^t, where $W \times H$ is equal to the size of input images I and C means the classes seen over time.

Furthermore, suppose that we have a training dataset D_{tr}, where $D_{\text{tr}} = D_{\text{tr}}^1 \cup D_{\text{tr}}^2 \cup \cdots \cup D_{\text{tr}}^T$ and $D_{\text{tr}}^{t-1} \cap D_{\text{tr}}^t = \emptyset$. For each step, $D_{\text{tr}}^t = \left(x_i^t,\ \hat{y}_i^t\right)|_{i=1}^{N^t}$, where N^t is the total number of images in step t. $x_i^t \in I^t$ is the ith input image with the size of $W \times H$ and I^t is the set of input images. $\hat{y}_i^t \in \mathcal{Y}^t$ is the one-hot segmentation ground-truth (GT) map corresponding to the image x_i^t and $\mathcal{Y}^t$ is the set of GT segmentation mask of task t. In addition to the real background class c_{bg}(only predicted in C^1), in this article, we suppose that the sets of annotations are disjoint at different learning steps, scilicet $\mathcal{Y}^{t-1} \cap \mathcal{Y}^t = \emptyset$. More specifically, $\mathcal{Y}^t$ only contains the annotations of current classes C^t. For all other labels, the classes that have been learned in the past $C^{1:t-1}$ or the classes to be learned in the future $C^{t+1:T}$ or c_{bg} are classified as C^0. M^t should have the ability to predict all the classes seen over time.

Furthermore, we describe the two concepts involved in the two main challenges of CISS in RS images as follows. First, CF, which means that M^t will dramatically forget the learned classes $C^{1:t-1}$ when learning C^t. The second problem is the shift of background class C^0. At learning step t, $C^0 = c_{\text{bg}} \cup C^{1:t-1} \cup C^{t+1:T}$, but at step $t+1$, $C^0 = c_{\text{bg}} \cup C^{1:t} \cup C^{t+2:T}$. In other words, the pixels belonging to C^0 change as the learning step changes. Therefore, the network may be so confused that may not be able to learn well. Table I shows the definition of symbols involved in the problem definition and their meaning in the proposed method.

B. Network Framework

The proposed CISS framework in RS images based on the historical information is shown in Fig. 4. Θ^{t-1} of the old model is frozen, while Θ^t is learnable. We adopt DeepLab-V3 [55] and ResNet101 [56] as the principal part of our network. L_1–L_4 express the four residual blocks of ResNet101, which is the body meanwhile the encoder of the framework.

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The output of the fourth residual block is the intermediate feature layer, as discussed in [37]. Atrous spatial pyramid pooling (ASPP) is the head and the decoder of the framework. Since the large-scale transformation of RS images, the application of the ASPP module can obtain the multiscale receptive field, which helps better learn the objects of different sizes. The combination of residual block and ASPP comprises the task-shared layers used by the continual learning network to learn high-dimensional semantic representations from input images. At the end of the framework, a task-specific layer is added to predict semantic labels. The task-specific layer consists of a classifier with the same number of 1×1 convolution as the incremental steps. We employ the softmax activation function to get the final output probability maps. During learning step t, training samples of D_{tr}^t will be fed into the old model M^{t-1} and the current model M^t as input. The network processes the input data through the encoding and decoding stages and ultimately obtains the predicted segmentation mask. Throughout the training process, Θ^{t-1} is frozen, while Θ^t is fully learnable. Classically, the CE loss is applied to learn the new classes, while the OPD loss is employed to avoid drastic performance degradation in the learned classes.

Generally, in the CIL setting, CE loss is defined as follows:

$$\mathcal{L}_{\text{ce}}\left(x_i^t, \hat{y}_i^t\right) = -\sum_{c \in C^t} \hat{y}_i^t[c] \cdot \log\left(\widehat{P}\left(x_i^t; \Theta^t\right)[c]\right) \tag{1}$$

where $\hat{y}_i^t[c]$ and $\widehat{P}\left(x_i^t; \Theta^t\right)[c]$ represent the one-hot GT and the output probability for class c, respectively.

Formally, a standard choice for OPD loss is

$$\mathcal{L}_{\text{opd}}\left(x_i^t, \hat{y}_i^t\right) = -\sum_{c \in \mathcal{Y}^{t-1}} \widehat{P}\left(x_i^t; \Theta^{t-1}\right)[c] \cdot \log\left(\widehat{P}\left(x_i^t; \Theta^t\right)[c]\right) \tag{2}$$

where $\mathcal{Y}^{t-1} = \{C^1, C^2, \ldots, C^{t-1}\}$ includes previously learned categories, Θ^{t-1} is the parameters set, and $\widehat{P}\left(x_i^t; \Theta^{t-1}\right)[c]$ is the output probability of M^{t-1} at step t.

Though the alliance of CE loss and OPD loss can achieve good results, we find that not considering the combination of background class processing with the historical information is detrimental to retaining the old knowledge since the existence of background shift problem. In addition, the intermediate layer of the feature space covers valuable high-level semantic representation information as confirmed in [37]. Taking [37] as the benchmark, our selection of intermediate feature layer is consistent with it. Therefore, enhancing the global perception of features is beneficial for preserving previous knowledge. On account of these cases, we propose the historical information-guided feature representation distillation (HFPD) loss to focus on global information about features, especially regions associated with the foreground of previous classes. We replace CE loss with the proposed historical information-guided cross-entropy (HCE) loss, which helps retrieve the old classes' labels from the background class.

Define $e_i^{t-1} \in \mathcal{R}^{H_f \times W_f \times C_f}$ and $e_i^t \in \mathcal{R}^{H_f \times W_f \times C_f}$ as the ith intermediate feature of step $t-1$ and step t, respectively. The overall loss of our framework can be expressed by the following formula:

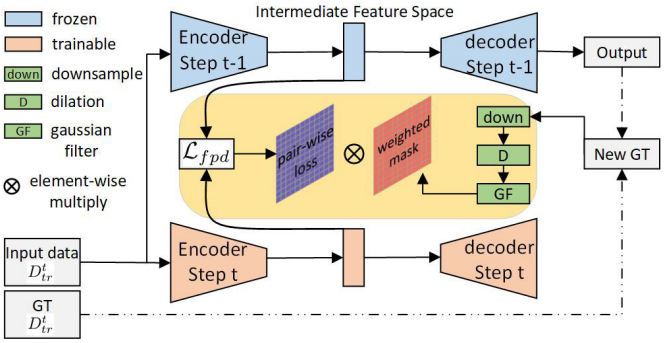


Fig. 5. Illustration of feature global perception module, which is shown in the yellow rounded rectangle area. Instead of pair-wise computation only for the intermediate feature between the old and new models, we add a global weighted mask to enhance the new model's perception of the object regions about the old model. The core of the proposed module is the orange grid weights generated by multiple green rectangles operation, which distributes different attention to various areas of the feature. Therefore, we can better preserve the significant information from the previously learned classes.

$$\mathcal{L}\left(\Theta^t\right) = \frac{1}{|N^t|}\sum_{i=1}^{N^t}\left(\mathcal{L}_{\text{hce}}\left(x_i^t, \hat{y}_i^t\right) + \lambda\mathcal{L}_{\text{opd}}\left(x_i^t, \hat{y}_i^t\right) + \beta\mathcal{L}_{\text{hfpd}}\left(e_i^{t-1}, e_i^t\right)\right) \tag{3}$$

where $\lambda, \beta > 0$ are hyperparameters balancing the importance of the three terms; H_f, W_f, and C_f denote the width, height, and channel of the intermediate feature, respectively; N^t is the total number of images; and x_i^t and $\hat{y}_i^t$ are the ith input image and the corresponding one-hot segmentation GT map of step t, respectively.

The pipeline of the historical information-guided CISS is shown in Algorithm 1.

C. HFGP Module

Keeping the internal stage close to the previous step in the $L2$ loss is beneficial to retaining the knowledge of the old model, as demonstrated in ILT [37]. The effectiveness and importance of adding constraints in the intermediate feature space are also proved in the existing distillation approaches [53], [57]–[59]. The distillation loss based on intermediate feature can be formulated as

$$\mathcal{L}_{\text{fpd}} = \frac{1}{N}\sum_{h=1}^{H_f}\sum_{w=1}^{W_f}\sum_{c=1}^{C_f}\mathcal{M}\left(e_{h,w,c}^{t-1} - e_{h,w,c}^t\right)^2 \tag{4}$$

where $N = H_f W_f C_f$ is the total number of elements in intermediate feature and $\mathcal{M}$ indicates the imitation mask, i.e., binary mask in [58] and fine-grained mask in [57]. As for the mean squared error (MSE) loss used in ILT [37], $\mathcal{M} = 1$ for all pixels of the intermediate feature. In other words, the distillation loss on the intermediate level of ILT [37] can be written as

$$\text{MSE} = \frac{1}{N}\sum_{h=1}^{H_f}\sum_{w=1}^{W_f}\sum_{c=1}^{C_f}\left(e_{h,w,c}^{t-1} - e_{h,w,c}^t\right)^2 \tag{5}$$

where $e_i^{t-1} \in \mathcal{R}^{H_f \times W_f \times C_f}$ and $e_i^t \in \mathcal{R}^{H_f \times W_f \times C_f}$ are the ith intermediate feature of step $t-1$ and step t, respectively.

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Algorithm 1 Continual Learning With Historical Information

Input:
$\mathcal{T} = 1, 2, \ldots, t, \ldots, T$: the CIL steps;
$D_{\text{tr}}^t = \left(x_i^t,\ \hat{y}_i^t\right)|_{i=1}^{N^t}$: data of trainig step t, x^t, $\hat{y}^t$ are images and the corresponding ground truths respectively;
λ, β: hyperparameters of different loss;
Output:
M^t, Θ^t: model and optimization parameters for step t;
while t **do**
  Obtain the training data D_{tr}^t of the current step;
  if t = 1 **then**
    Load the pretrained weights and initialize Θ^1;
    Exploit Θ^1 on the training data D_{tr}^1: $\widehat{P}\left(x_i^1; \Theta^1\right)$;
    Calculate the CE loss: $\mathcal{L}_{\text{ce}}\left(x_i^1, \hat{y}_i^1\right)$ (1);
    Update Θ^1: $\Theta^1 = \text{argmin}_{\{\theta_i^1\}}\mathcal{L}_{\text{ce}}$;
    Save model M^1;
  else
    Load Θ^{t-1} to initialize Θ^t;
    Exploit Θ^{t-1} on the training data D_{tr}^t: $\widehat{P}\left(x_i^t; \Theta^{t-1}\right)$;
    Exploit Θ^t on the training data D_{tr}^t: $\widehat{P}\left(x_i^t; \Theta^t\right)$;
    Get Pseudo-labels: $\bar{y}_i^t$ (13);
    Calculate the HCE loss: $\mathcal{L}_{\text{hce}}\left(x_i^t, \bar{y}_i^t\right)$ (14);
    Calculate HFPD loss $\mathcal{L}_{hfpd}(x_i^t)$ between the end of the body θ_e^{t-1} and θ_e^t (11);
    Calculate OPD loss between $\widehat{P}\left(x_i^t; \Theta^{t-1}\right)$ and $\widehat{P}\left(x_i^t; \Theta^t\right)$ (2);
    Integrate all losses: $\mathcal{L}(\Theta^t)$ (3);
    Update Θ^t: $\Theta^t = \text{argmin}_{\{\theta_i^t\}}\mathcal{L}(\Theta^t)$;
    Save model M^t;
  end if
end while
return M^t, Θ^t

However, there are some deficiencies in calculating the $L2$ distance directly without distinguishing the importance of each pixel in the intermediate feature space. On the one hand, the treatment of all intermediate feature pixels of M^t equally does not make the model pay more attention to the object regions learned by the old model, which is not conducive to correctly retaining the valuable knowledge of the old model. In addition, the ignorance of the object regions further deteriorates the imbalance between positive and negative samples of the training data. On the other hand, it also restricts the ability to learn new knowledge for the feature pixels that are not significant to retain the learned model knowledge.

Based on the above analysis, we add a weighted mask $\mathcal{M}_w$ to the intermediate feature space, which focuses on the foreground region of the old classes. As mentioned before, for the pixels potentially belonging to the learned classes, $\mathcal{M}_w$ distributes a considerable weight, while irrelevant pixels obtain a tiny weight. This strategy not only ensures that the new model remembers the knowledge of the old model as much as possible but also can learn new knowledge.

We first fabricate a binary mask M_{bg} for the segmentation GT map of step t to determine whether each pixel is a background, which can be expressed as

$$M_{\text{bg}} = \mathbb{I}\left(\hat{y}_i^t(w, h) \in C^0\right) \tag{6}$$

where $\mathbb{I}$ denotes the indicator function. In other words, pixels labeled as background class have a value of 1, while the rest have a value of 0. Naturally, the weighted mask $\mathcal{M}_w$ can be generated by the following formula:

$$\mathcal{M}_w(w, h) = \begin{cases} 1, & \text{if } M_{\text{bg}} = 1 \ \&\ \text{argmax}(\widehat{P}^{t-1})! = 0 \ \&\ T \\ 0, & \text{otherwise} \end{cases} \tag{7}$$

where T is the condition that capacitates the selected pixel relatively confident. However, $\mathcal{M}_w$ obtained in this way only allows the feature pixels that the network judges to be the foreground region to participate in the backpropagation of the gradient. It is not conducive to learning new classes if feature pixels with little correlation with the learned classes have no opportunity to join in the loss calculation.

In consideration of this, we employ the Gaussian Blur with a Gaussian kernel to smooth $\mathcal{M}_w$. However, the feature map obtained after the processing of the coding stage has been reduced several times than the original input image. Therefore, the number of values of 1 attained in formula (7) is particularly few. Even though the Gaussian Blur is exploited for smoothing, the weight of most feature pixels is still valued of 0. In this instance, the whole feature space focuses on extremely few locations, and the feature pixels involved in loss calculation are also very rare, which is adverse to retaining the old knowledge and learning new classes.

Based on the above analysis, we exploit a dilation operation to expand the number of 1 value in formula (7) before Gaussian Blur to make the foreground region concerned by the model larger. We apply the softmax activation function after the Gaussian smoothing to get our final $\mathcal{M}_w$, which can be denoted as

$$\mathcal{M}_w = \frac{1}{N}\sum_{h=1}^{H_f}\sum_{w=1}^{W_f}\sigma\left(\text{Gauss}(D(\mathcal{M}_w(w, h))\right) \tag{8}$$

where σ is on behalf of the softmax activation function, $Gauss$ represents the Gaussian Blur, while D indicates the dilation operation, and $\mathcal{M}_w(w, h)$ is produced by formula (7). Note that, the softmax function used here just for normalization.

Eventually, HFPD loss can be expressed as

$$\mathcal{L}_{\text{hfpd}}(x_i^t) = \frac{1}{N_t}\sum_{i=1}^{N^t}\mathcal{M}_w \odot \text{MSE}\left(e_i^{t-1}, e_i^t\right). \tag{9}$$

Note that MSE is the loss between the embedding vectors of the old and new models. As shown in Fig. 5, the weighted mask obtained through the above steps is multiplied by the pair-wise calculation of the intermediate feature, which allows extra attention to be assigned to specific regions. Therefore, the proposed HFPD loss helps achieve a tradeoff between elasticity and rigidity.

D. HLR Module

During the learning step t, the input images only have annotations of the background class C^0 and the current classes C^t. As mentioned in Section I, background class $C^0 = c_{\text{bg}} \cup C^{1:t-1} \cup C^{t+1:T}$ carries historical and future information. If we

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leave these pixels untreated, some important knowledge will be lost. Since we cannot predict in advance what knowledge we will learn in the future, what we can only utilize is the parameter information that the old model provides. The LR module is inspired by the effectiveness of pseudo-labels in the field of domain adaptation [60]. We employ the output information of the old model $\widehat{P}\left(x^t;\Theta^{t-1}\right)$ to generate pseudo-labels about the old classes. This strategy contributes to enriching the annotation information of the current step, thereby reducing the CF of the learned classes.

We define $\bar{y}^t \in \mathcal{R}^{W\times H\times C}$ as the target of the segmentation GT map obtained through our pseudo-labels strategy at step t, where $W \times H$ indicates the input image size and C denotes the category. Similar to the HFPD loss, we first acquire a binary mask $\mathcal{M}_{\text{bg}}$ indicating whether the GT label is a background class according to formula (6). Then, we obtain the output probability of the old model on the current sample $\widehat{P}^{t-1}$. Ultimately, the ith pseudo-labels extracted from step t can be expressed as

$$\bar{y}_i^t(w,h,c)=\begin{cases}1, & \text{if } \mathcal{M}_{\text{bg}}=0 \text{ and } c=\underset{c'\in\mathcal{C}^t}{\text{argmax}}\hat{y}_i^t\left(w,h,c'\right)\\ 1, & \text{if } \mathcal{M}_{\text{bg}}=1 \text{ and } c=\underset{c'\in\mathcal{C}^{1:t-1}}{\text{argmax}}\widehat{P}_i^{t-1}\left(w,h,c'\right)\\ 0, & \text{otherwise.}\end{cases} \tag{10}$$

In other words, in the case of pixel (w,h) belonging to the nonbackground class, we reserve the GT label; otherwise, if $\widehat{P}^{t-1}$ determines the pixel pertains to be a specific class in the old classes, it will be substituted for the corresponding class label index. However, since the limitation of the ability of the learned old model itself, the pixels judged as the old classes may not be correct, and the pseudo-labels selected by the above formula are in low confidence level. Therefore, we utilize the pseudo-labels generation method in [61] for reference and optimize the above formula as follows:

$$\bar{y}_i^t(w,h,c)=\begin{cases}1, & \text{if } \mathcal{M}_{\text{bg}}=0 \text{ and } c=\underset{c'\in\mathcal{C}^t}{\text{argmax}}\hat{y}_i^t\left(w,h,c'\right)\\ 1, & \text{if } \mathcal{M}_{\text{bg}}=1 \text{ and } c=\underset{c'\in\mathcal{C}^{1:t-1}}{\text{argmax}}\widehat{P}_i^{t-1}\left(w,h,c'\right)\\ & \text{and top1-top2}>\tau\\ 0, & \text{otherwise}\end{cases} \tag{11}$$

where top1 and top2, respectively, correspond to the maximum and submaximum values of the $\widehat{P}^{t-1}$ except the background class and τ is the threshold of the difference between top1 and top2 values. When the difference is larger than the threshold τ, the pseudo-labels are considered of sufficient confidence to be preserved. The CE loss with our historical information guided label construction of the old classes can be represented as

$$\mathcal{L}_{\text{hce}}\left(x_i^t,\bar{y}_i^t\right)=\sum_{c\in\mathcal{C}^t}\bar{y}_i^t(w,h,c)\log\widehat{P}_i^t(w,h,c) \tag{12}$$

where $\bar{y}_i^t(w,h,c)$ is obtained by formula (13) and $\widehat{P}_i^t(w,h,c)$ is the output probability of the current model about class c. The proposed LR module can be seen in Fig. 6. We can observe

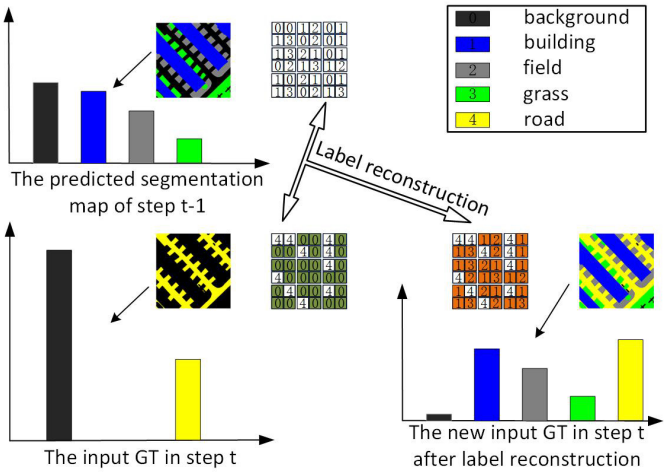


Fig. 6. Illustration of LR module. We combine the output of the old model on the current data (top left) with the input GT of the current step (bottom left) to obtain the pseudo-labels (middle-right) of the new input GT on the current step.

that a new label more consistent with the labeled map can be obtained by combining the historical information in the old model with the GT input currently. Therefore, this module effectively improves the CSS background class shift problem.

IV. Experiments

In this section, we first introduce experimental settings in Section IV-A. Then, we briefly present our baselines in Section IV-B. In Sections IV-C and IV-E, we conduct and analyze ablation studies on iSAID and GCSS datasets, respectively. Finally, we analyze a large number of experiments on the two datasets to verify the effectiveness of the proposed framework in the rest sections.

A. Datasets, Metrics, and Implementation Details

1) iSAID [44]: The iSAID dataset is captured by Gao Fen (GF)-1 and GF-2 satellites and contains 2806 RS images. It is widely used in the field of semantic segmentation and consists of 1411 images in the training set, 458 images in the validation set, and 957 images in the test set with a wide range of sizes. These images' annotations contain a background category and 15 object categories with 655 451 instances in total. The categories are listed in alphabetic as follows: Background, Baseball Diamond, Basketball Court, Bridge, Ground Track Field, Harbor, Helicopter, Large Vehicle, Plane, Roundabout, Ship, Small Vehicle, Soccer Ball Field, Storage Tank, Swimming Pool, and Tennis Court. Since the labels in the test set are not available, 20% of the training set is used for validation, and the validation set is used for the test.

2) GCSS [45], [46]: The GCSS dataset is captured by the GF-2 satellite and contains 948 red, green, blue color mode (RGB) images with spatial resolution better than 1 m. It is a semantic segmentation dataset from the 2020 GF Challenge, and the size of all images is of 2000 × 2000 pixels. We use 760 images as the training set and perform inference on the other 188. The annotations contain eight object categories

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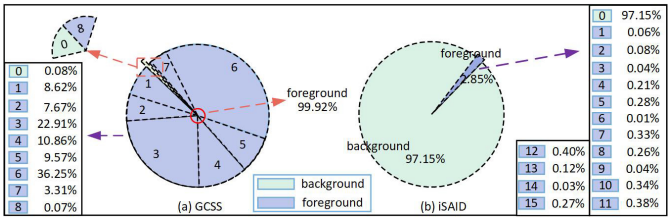


Fig. 7. Comparison of the foreground and background pixels ratio of the two datasets used. The GCSS dataset almost only has the foreground category, while the other background category accounts for most of the proportion. The numbers in the left and right represent the alphabetical categories in GCSS. Similarly, the numbers under the bar chart correspond to the alphabetical categories in the iSAID dataset. These two datasets with large differences between the foreground and the background proportion verify the generalization of our model.

and one other class. The categories in alphabetical order are Background (Other), Building, Field, Grass, Impervious Surfaces, Road, Tree, Vehicle, and Water.

As shown in Fig. 7, there is a noteworthy distinction between the background proportions of the two datasets. This helps to test the generalization of different methods on datasets with a huge gap between the foreground and the background.

3) Metrics: We employ the mean intersection over union (mIoU) [62] as well as the backward transfer (BWT) [20] to fairly evaluate the performance of all methods. Assume that $\text{mIoU}_{t,k}(k \leq t)$ is the accuracy calculated on the test set of the kth step after learning tasks from 1 to t, which can be expressed as follows:

$$\text{mIoU}_{t,k} = \frac{1}{C^k} \sum_{i=1}^{C^k} \frac{p_{ii}}{\sum_{j=1}^{C^k} p_{ij} + \sum_{j=1}^{C^k} p_{ji} - p_{ii}} \tag{13}$$

where C^k indicates the number of categories in the kth step and p_{ij} represents the number of pixels belong to class i while predicting to class j. Then, the BWT is introduced to reflect the memory capacity of different continual learning methods. Note that this value is usually negative due to CF. In particular, the BWT up to task t is defined as

$$\text{BWT} = \frac{1}{t-1} \sum_{k=1}^{t-1} (R_{t,k} - R_{k,k}) \tag{14}$$

where $R_{t,k}$ represents the test accuracy on task k of the model obtained after learning task t. The larger the two evaluation metrics, the better the model.

4) Implementation Details: All experiments are conducted on a single NVIDIA Tesla V100 graphics processing unit (GPU) with 32-GB memory. Also, all methods are implemented with Pytorch and tested under the same experimental settings. In particular, following the experimental setup mentioned in [38], we use the Deeplab-V3 architecture [55] with an output stride of 16 and adopt ResNet101 [56] as the backbone for all approaches. ImageNet pretrained model [63] has been employed to initialize the backbone in the first learning step. We apply stochastic gradient descent with a polynomial decay rule with power 0.9 to train the network. For all methods, our initial learning rate is set to 0.01 in the first learning step and 0.001 for all subsequent steps. For the iSAID dataset, we crop the images into a size of 896 × 896 pixels with the overlap of the sliding window being 384. We execute the same processing as iSAID for the images of the GCSS dataset. We train the model 30 epochs for GCSS and 40 epochs for iSAID with a batch size of 24 in every learning step. The training images (all resized to 512 × 512 pixels) are randomly cropped and flipped to enhance the samples. We establish the incremental datasets D_{tr}^t by dividing the entire dataset into disjoint image sets. There are no constraints other than ensuring that the classes in the C^t contain a minimum number of images with labeled pixels.

B. Baselines

Since segmentation is considered a pixel-level classification problem, we compare our method with standard continual learning benchmarks that are initially designed for classification tasks. In particular, we report the results of seven regularization-based approaches (three prior-focused and four data-focused) on iSAID and four data-focused methods on GCSS. In addition, we conduct experiments on the other two baselines for both datasets: FT and offline learning (Joint). Joint is often regarded as the performance upper bound of incremental learning since it trains all data offline at once.

For the prior-focused methods, we are consistent with the setup of MiB [38] and select these three methods: EWC [34], SI [42], and RW [16], while LWF [35], iCaRL [19], ILT [37], and MiB [38] are chosen for the data-focused approaches. LWF is the ancestor of data-focused methods and it is the first work to apply the KD strategy to incremental learning. The original CE $\mathcal{L}_{\text{ce}}$ and distillation $\mathcal{L}_{\text{opd}}$ loss shown in formulas (1) and (2) refer to the two losses defined in LWF. LWF-multi-class (MC) is the single-head version of LWF and is based on multiple binary classifiers. ILT and MiB are specifically proposed for CSS. MiB ameliorates the two loss functions of LWF by modeling the background. ILT has two distillation losses: one is used in the output space and the other is used in the feature space. Followed by ILT, our framework applies these two distillation losses and CE loss. However, we adopt the variation of feature space distillation loss and CE loss guided by historical information.

C. Ablation Study on the iSAID Dataset

To the best of our knowledge, we are the first to study CIL in semantic segmentation of RS images without replaying the data of the previous classes. Therefore, there is no reference partition of incremental settings on the iSAID dataset. Learning from the division of commonly used segmentation datasets by methods [37], [38], we perform four incremental settings for the iSAID as follows: 14-1, 10-5, 5s, and 10-1s. These settings refer to learn 14 then 1 class (two steps), 10 then 5 classes (two steps), 5 classes per learning step multiplied by 3 (three steps), and 10 classes followed by five times 1 class (six steps). It is worth noting that all incremental categories are in the alphabetical order. Table V shows quantitative experiments of the iSAID in 14-1, 10-5, 5s, and 10-1s.

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TABLE II
PARAMETER ANALYSIS OF THE PSEUDO-LABEL THRESHOLD τ ON THE iSAID TEST SET

Method	**14-1** (2 steps)			**10-5** (2 steps)			**5s** (3 steps)				**10-1s** (6 steps)		
	1-14	15	all	1-10	10-15	all	1-5	5-10	11-15	all	1-10	11-15	all
ILT [37]	41.00	59.67	42.24	49.25	35.14	44.55	40.80	9.68	31.30	27.26	15.58	19.46	16.87
τ = 0.3	44.91	60.05	45.92	52.79	33.22	**46.27**	43.51	11.76	30.48	28.59	28.97	23.06	27.60
τ = 0.4	45.74	57.04	**46.49**	52.81	33.15	46.24	43.37	13.42	31.09	29.29	30.46	22.32	**27.75**
τ = 0.5	45.20	58.62	46.09	52.51	33.25	46.09	44.43	12.94	30.81	**29.39**	30.25	22.63	27.71
τ = 0.6	45.57	58.74	46.45	52.13	33.19	45.82	42.90	13.02	31.17	29.03	28.58	25.75	27.64

TABLE III
PERFORMANCE COMPARISONS OF MEAN IOU ON THE iSAID TEST SET FOR DIFFERENT MASK GENERATION METHODS

Method	**14-1** (2 steps)			**10-5** (2 steps)			**5s** (3 steps)				**10-1s** (6 steps)		
	1-14	15	all	1-10	10-15	all	1-5	5-10	11-15	all	1-10	11-15	all
ILT [37]	41.00	59.67	42.24	49.25	35.14	44.55	40.80	9.68	31.30	27.26	15.58	19.46	16.87
$T_I(fg)$	47.41	63.40	48.48	34.42	14.61	27.82	19.80	7.49	21.93	16.41	10.70	11.54	10.98
$T_D(fg)$	48.03	60.52	48.87	34.98	24.88	31.62	18.77	14.16	16.61	16.51	16.29	15.18	15.92
T_I	**48.36**	56.69	48.92	**53.61**	38.43	**48.55**	**44.68**	14.74	31.09	30.17	**31.97**	24.00	29.32
T_D	48.06	**70.40**	**49.55**	52.52	**39.52**	48.19	43.54	**23.94**	**35.67**	**34.39**	31.48	**27.65**	**30.21**

1) Sensitivity Analysis of the Confidence Threshold τ: In [61], the guidance threshold for pseudo-label generation is 0.4 and 0.5 in two different tasks. Therefore, we conduct four groups of experiments to explore what value of τ can achieve the best results on the iSAID dataset. From Table II, we can observe that the value of τ varies for different incremental settings. On the 5s setting, τ is set as 0.5 achieving the best performance, while τ is 0.3 on the 10-5 setting. On the setting of 14-1 and 10-1s, τ is set as 0.4 works best. However, the overall effect is the most prominent under the threshold τ set as 0.4. Therefore, we apply threshold τ set as 0.4 for all incremental settings.

2) Evaluation of Different Weighted Masks $\mathcal{M}_w$: We adopt two different forms for the condition T to obtain the binary $\mathcal{M}_w(w,h)$ of foreground region. One is that $T = \widehat{P}^{t-1}$>0.5, and the other is $T = top1-top2 > \tau$ just as formula (11). The discrepancy between the two forms lies in that $\mathcal{M}_w$ generated by the former is completely independent and not affected by the generation of pseudo-label, while the latter depends on the generation of the pseudo-label. We list four versions of the final $\mathcal{M}_w$.

1) *T_I(fg):* $T_I(fg)$ represents that the final $\mathcal{M}_w$ is obtained by the equation independent of the pseudo-label without the dilation operation and Gaussian smoothing processing. In other words, this is a binary $\mathcal{M}_w$ acquired by the first form of (7) and only focuses on the object region of the input image.
2) *T_D(fg):* Contrast to $T_I(fg)$, the only difference between them is that $T_D(fg)$ gains the final binary $\mathcal{M}_w$ through pseudo-label.
3) *T_I:* T_I means that the final $\mathcal{M}_w$ is independent of the generation of pseudo-label, and it will be processed by dilation and Gaussian blur. Therefore, we get a soft weighted mask through this processing.
4) *T_D:* Compared with T_I, T_D exploits the pseudo-label strategy to identify the old classes from the current background pixels. Then, it smoothes the binary $\mathcal{M}_w$ to obtain the final version.

The results of different weighted masks $\mathcal{M}_w$ are shown in Table III. The binary $\mathcal{M}_w$ of the foreground region generated by using the pseudo-label is 2% points(*p.p*) higher than the method without the pseudo-label on average. Similarly, it can be observed that the pseudo-label approach is still better for the generation of soft $\mathcal{M}_w$. In addition, the average mIoU of the soft $\mathcal{M}_w$ is nearly 20% higher through the expansion of the object region and Gaussian smoothing operations on the obtained binary $\mathcal{M}_w$ except for the setting of 14-1. The overall results of the 14-1 setting are not much different, and all are above 48% and below 50%. We consider that this is because the learned classes have accounted for 93% of the total target classes, and the object region has much more pixels than other settings. Therefore, different forms of $\mathcal{M}_w$ have little influence on the results. The results in Table III prove the effectiveness of the proposed HLR strategy and the feature global perception module. The subsequent reported experimental results are all carried out on $\mathcal{M}_w$ generated by **T_D**.

3) Performance Comparison of Different Components for CIL Tasks: We report the detailed contributions of different components in Table IV. First, we conduct experiments on LWF [35], which is the first method to introduce the

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TABLE IV
COMPARISONS OF DIFFERENT COMPONENTS ON THE iSAID TEST SET

OPD	CE	FPD	HCE	HFPD	**14-1** (2 steps)			**10-5** (2 steps)			**5s** (3 steps)				**10-1s** (6 steps)		
					1-14	15	all	1-10	10-15	all	1-5	5-10	11-15	all	1-10	11-15	all
√	√				39.57	67.84	41.46	44.70	45.41	44.94	36.53	26.70	38.44	33.89	4.99	16.81	8.93
√	√	√			41.00	59.67	42.24	49.25	35.14	44.55	40.80	9.68	31.30	27.26	15.58	19.46	16.87
√		√	√		45.74	57.04	46.49	52.81	33.15	46.24	43.37	13.42	31.09	29.29	30.46	22.32	27.75
√			√	√	48.06	70.40	**49.55**	52.52	39.52	**48.19**	43.54	23.94	35.67	**34.39**	31.48	27.65	**30.21**

TABLE V
MEAN IOU ON THE iSAID TEST SET FOR DIFFERENT CIL METHODS

Method	**14-1** (2 steps)			**10-5** (2 steps)			**5s** (3 steps)				**10-1s** (6 steps)		
	1-14	15	all	1-10	10-15	all	1-5	5-10	11-15	all	1-10	11-15	all
FT	0.32	**84.67**	5.94	0.00	**51.10**	17.03	0.00	0.00	**47.64**	15.88	0.00	15.63	5.21
SI [42]	0.28	84.47	5.89	0.00	50.64	16.88	0.00	0.00	47.61	15.87	0.00	15.59	5.20
EWC [34]	3.58	71.99	8.14	1.31	36.10	12.90	0.01	2.00	22.27	8.09	0.00	12.01	4.00
RW [16]	2.27	76.41	7.21	0.46	38.80	13.27	0.00	1.89	23.78	8.56	0.00	11.44	3.81
LWF [35]	39.57	67.84	41.46	44.70	45.41	44.94	36.53	**26.70**	38.44	33.89	4.99	16.81	8.93
iCaRL [19]	41.98	39.54	41.81	50.96	24.11	42.01	42.08	5.20	23.00	23.42	12.19	6.09	10.16
ILT [37]	41.00	59.67	42.24	49.25	35.14	44.55	40.80	9.68	31.30	27.26	15.58	19.46	16.87
MiB [38]	**48.38**	81.96	**50.62**	47.80	47.34	47.65	**46.75**	17.83	37.54	34.04	25.06	22.94	24.35
Ours	48.06	70.40	49.55	**52.52**	39.52	**48.19**	43.54	23.94	35.67	**34.39**	**31.48**	**27.65**	**30.21**
Joint	56.58	85.37	58.50	58.81	57.88	58.50	57.63	59.98	57.88	58.50	58.81	57.88	58.50

KD strategy into continual learning. It only uses the OPD and the CE loss. Then, we show the performance of ILT [37], which not only applies distillation loss to the output layer (OPD) but also adds constraints to the intermediate feature layer [feature representation distillation (FPD)] before decoding. We employ the HCE loss via historical information-guided pseudo-label strategy to substitute for the ordinary CE in the ILT. The results are shown in the third row of Table IV. Compared with ILT, the performance in the four incremental settings has been significantly improved, from left to right by +4.25, +1.69, +2.03, and +10.88 *p.p*. Furthermore, the best performance is achieved after replacing the FPD loss in ILT with the HFPD loss. Our method outperforms ILT by a wide margin both on old and new classes, especially on the long sequence setting 10-1s (six steps). These results confirm the importance and validity of taking full advantage of historical information.

D. Quantitative Analysis on the iSAID Dataset

1) Addition of One Class(14-1): In this experimental setup, we carry out two learning steps: the first we learn is the top 14 classes, then followed by tennis court class. From the first row of Table V, we can recognize that FT the network can result in significant degradation of the performance with a final mIoU of 5.94%. Though FT performs excellently in the latest step, it almost completely forgets the old classes. This demonstrates that even if only one class is appended, CF can occur in the semantic segmentation scenarios. Unexpectedly, using the SI regularization strategy does not improve the performance, while RW and EWC only increase benefits by an average of nearly 2 (*p.p*). Compared with prior-focused methods (SI, EWC, and RW), data-focused approaches are vastly superior to them, with mIoU higher than 30 *p.p* on average. In fact, LWF, iCaRL, ILT, and MiB outperform them by a large margin in the first step, which demonstrates that this approach can effectively prevent CF. While ILT exceeds standard CIL baselines, we can obtain further improvement. Compared with ILT, we gain 7.06 and 10.73 *p.p* in retention of previous knowledge and learning of new classes, respectively. The results illustrate the importance of using historical information in CIL of semantic segmentation for RS images. The proposed method shows comparable performance with MiB in this setting.

2) Addition of Five Classes in Single-Step(10-5): In this experiment, the following classes, small vehicle, soccer ball field, storage tank, swimming pool, and tennis court, are added in a single step after the first step. As can be seen from Table V, the overall performance is consistent with the 14-1 setting: FT and SI do great in the current step, but they nearly forget the task they have learned before. EWC and RW can neither retain old knowledge nor learn novel classes commendably. LWF, iCaRL, and ILT surpass EWC and RW by far, while MiB extra elevates mIoU by 3 *p.p*. The overall performance of the proposed method is the best, exceeding MiB with 0.54 *p.p*

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TABLE VI
PARAMETER ANALYSIS OF THE PSEUDO-LABEL THRESHOLD τ ON THE TEST SET

Method	**7-1** (2 steps)			**4-2s** (3 steps)				**4-1s** (5 steps)		
	1-7	8	all	1-4	5-6	7-8	all	1-4	5-8	all
ILT [37]	67.10	77.60	68.41	64.77	76.60	44.31	62.61	61.73	59.41	60.57
τ = 0.3	68.04	77.10	**69.17**	65.82	78.01	44.74	**63.60**	64.26	59.53	**61.90**
τ = 0.4	68.03	77.12	**69.17**	65.49	77.65	44.77	63.35	64.19	59.56	61.88
τ = 0.5	67.69	74.96	68.60	65.27	77.80	44.41	63.19	64.33	59.44	61.89
τ = 0.6	67.73	74.62	68.60	65.43	77.70	44.72	63.32	64.25	59.41	61.83

on mIoU. Remarkably, we achieve the best results on the first step higher than the best baseline of 1.46 *p.p* and higher than MiB with 4.72 *p.p*. Besides, we obtain performance closer to the upper bound of joint training at the first step. This manifests that using historical information to reconstruct labels and enhance the global perception of features is beneficial to further mitigate CF.

3) Addition of Five Classes in Two Steps(5s): On 5s, we analyze the performance on three sequential steps of five classes. FT and prior-focused approaches behave poorly on the first two steps as they do in other settings. Data-focused methods implement different tradeoffs between preserving old classes and learning new ones. ILT and iCaRL both obtain a good result in steps 1 and 3, but they suffer from serious forgetting in step 2, which leads to their results in less than 30% in the end. LWF, MiB, and our method achieve good performance on both old and new classes. In particular, they are an average of 40 *p.p* higher than SI, EWC, and RW in the first step and an average of 20 *p.p* higher in the second step. Our approach is comparable to the best baseline performance in this setting.

4) Addition of Five Classes in Multistep(10-1s): This setup is similar to 10-5, except that the last five categories are learned sequentially, one after the other; 10-1s is the most challenging of the four settings because it contains the most incremental steps, six in total. From Table V, we can discover that FT and prior-focused strategies work poorly on both old and new classes with mIoU inferior to 6%. Even worse, they are incapable of avoiding CF, with a performance of nearly 0% in the first step (ten classes). Although the data-focused methods (LWF, iCaRL, ILT, and MiB) reduce the forgetting of old classes to a certain extent, they still suffer a severe loss in performance. Compared with adding five classes in a single step (10-5 setup), the performance of the multistep scenarios on all classes declines on average by more than 30%. While MiB surpasses ILT with the best performance among data-focused approaches by a wide margin: +7.48 *p.p* on all classes, +9.48 *p.p* on old classes, and +3.48 *p.p* on new classes, our framework is capable of achieving a better boost. The average mIoU is improved by 24.07% compared to MiB, and our method outperforms all baselines in both old (31.48%) and new (27.65%) classes. The experimental result indicates that our method achieves a good balance between learning new knowledge and retaining old knowledge even in the setting of long sequence incremental tasks.

5) Analysis of BWT Metric on the iSAID Dataset: We calculate the BWT scores after the model learned the last task for different incremental settings. Since the existence of CF, the BWT values are usually negative, which we can also observe from Fig. 8. Because the larger the value of BWT, the better the memory ability of the model, which corresponds to the low forgetting degree. As can be seen from Fig. 8, both FT and prior-based methods (SI, EWC, and RW) obtain small BWT values at different increment settings, indicating that these methods suffer from severe CF, which is consistent with what we observe under the mIoU metric. Furthermore, we can notice that although MiB outperforms ILT in the mIoU metric on all settings, from the BWT metric, the memory capability of MiB is inferior to ILT on the 10-1s setting and even worse than LWF and iCaRL. This demonstrates that only evaluating all methods with mIoU has certain limitations. It is worth noting that our method has optimal and much higher BWT scores on all three settings except for the setting of 14-1, which is slightly lower than MiB. This indicates that our two proposed modules based on historical information are very effective for retaining previous knowledge.

E. Ablation Study on the GCSS Dataset

Similarly, we perform three incremental settings for GCSS: 7-1, 4-2s, and 4-1s. GCSS 7-1 denotes that we learn seven classes first and then followed by one class (two steps) and 4-1s means learning four classes followed by four classes sequentially (five steps). The order of learning categories still refers to alphabetical order because the experimental results on the iSAID dataset explicate that the prior-focused methods have little effect on performance improvement. Therefore, we only contrast our method with the data-focused approaches (LWF, iCaRL, ILT, and MiB) on this dataset.

1) Sensitivity Analysis of the Confidence Threshold τ: Consistent with the iSAID dataset, we perform four experiments to obtain the optimal threshold τ for the generation of the pseudo-labels. The results of the experiment on the three settings are shown in Table VI. Different from iSAID, the optimal threshold is distinctive in different settings, the performance of GCSS is best in all three of the divided incremental setups under τ set as 0.3.

2) Effect of LR Module and HFGP Module: Figs. 9 and 10 quantitatively and qualitatively show the effectiveness of the two proposed modules. Note that the visual effect of the iSAID dataset is poor due to the 97% background pixels, so we only

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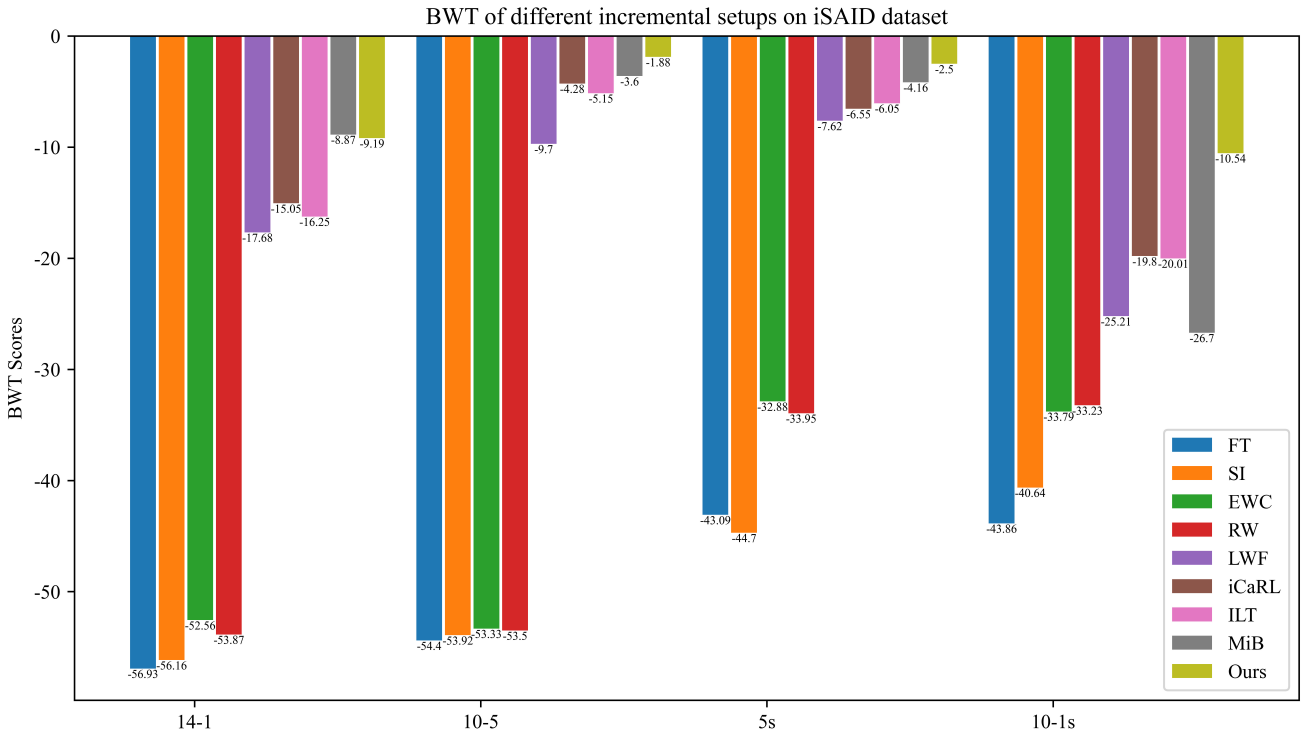


Fig. 8. BWT scores of different methods on the iSAID dataset with different incremental settings.

show the visualization on GCSS. As shown in Fig. 9, whether using LR modules or a combination of LR and HFGP modules, the learned classes (except for class 7) IoU obtained are higher than ILT and MiB. Fig. 10 shows the validity of the two proposed modules at the feature and output levels. From the feature maps (b) and (c), we can observe that by replacing the CE loss with the HCE loss associated with the proposed LR module, the noise existing on the feature maps is reduced. The combination of LR and HFGP further alleviates the noise on the feature maps as shown in column (d). From the predicted results, we can also discover that our method is more precise and accurate than the baseline.

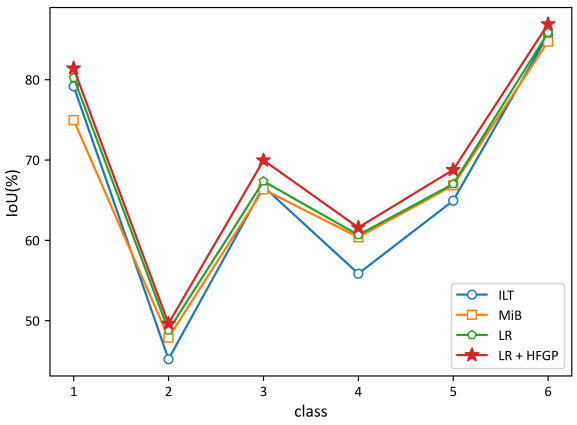


Fig. 9. Two proposed modules compare the performance of old classes with ILT and MiB methods in the 4-1s setting of the GCSS dataset. The seventh category is not shown in the figure because all methods have the same performance (all 0).

F. Quantitative Analysis on the GCSS Dataset

1) Addition of One Class(7-1): Our first CIL task is to add only one class in a single step. The experimental results are shown in Table VII. Unsurprisingly, FT is a bad strategy because it completely forgets the old task knowledge. In the old classes, LWF attains 41.26%, while iCaRL 56.14%, MiB 65.51%, and ILT 67.10%, which demonstrates the effectiveness of KD in preventing CF. Among these methods, ILT is significantly higher than others with mIoU that is 68.41%, only 3.05 *p.p* less than joint training. Compared with ILT, our method focuses more on constraining that the foreground pixels of the old and the new model are similar in the intermediate feature layer by distributing a large weight to the foreground pixels. Besides, we exploit pseudo-label to solve the background shift problem in the output layer. Therefore, our prediction outperforms all baselines both in the old classes (+1.06 *p.p*) and the overall performance (+1.01 *p.p*) compared to ILT. The experimental results demonstrate that our method improves the robustness of CF and background shift.

2) Addition of Two Classes in Two Steps(4-2s): In this experiment, we analyze the results of the 4-2s setting, which means learning 4 classes first followed by two sequential steps of 2 classes. In Table VII, we report the results of all classes after the last learning step. We can discover that FT works poorly for this setting as expected, and it entirely forgets the knowledge it has learned before. Although the performance of LWF is much better than FT, it still suffers from severe forgetting with mIoU only about 27%. iCaRL has excellent performance on the first step and stable performance on the

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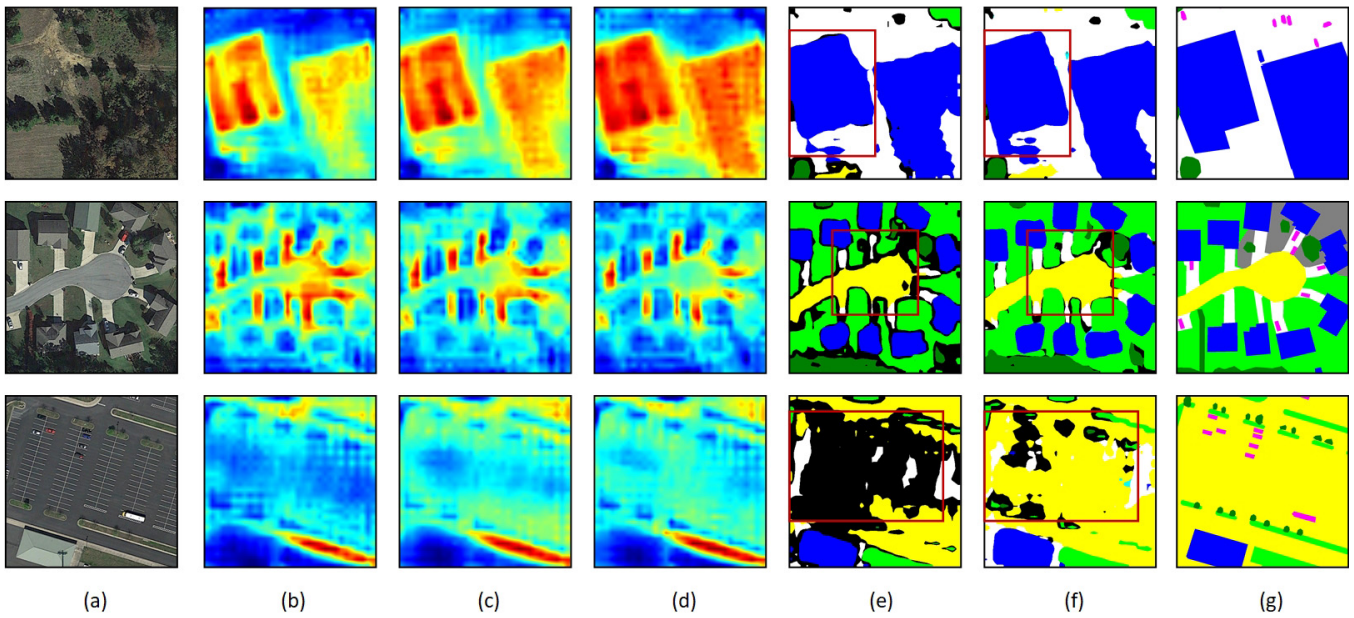


Fig. 10. Visualization of the two proposed LR and HFGP modules on the 4-1s (5 steps) setting of the Gaofen dataset. Note that in the 4-1s setting, the last step only contains the category of water. Different columns of pictures from left to right indicate: (a) input image, (b) feature maps for baseline (ILT [37]), (c) feature maps for LR, (d) feature maps for LR and HFGP, (e) predicted results of baseline (ILT [37]), (f) predicted results with LR and HFGP, and (g) GT. We choose three learned categories for visualization, including building in the first row, impervious surface in the second row, and road in the third row.

TABLE VII
MEAN IOU ON THE GCSS DATASET FOR DIFFERENT CIL METHODS

Method	**7-1** (2 steps)			**4-2s** (3 steps)				**4-1s** (5 steps)		
	1-7	8	all	1-4	5-6	7-8	all	1-4	5-8	all
FT	0.00	**88.00**	11.00	0.00	0.00	48.49	12.12	0.00	22.42	11.21
LWF [35]	41.26	87.61	47.06	3.46	58.70	45.57	27.80	0.00	22.18	11.09
iCaRL [19]	56.14	86.79	59.97	66.96	32.79	44.33	52.76	62.10	21.30	41.70
ILT [37]	67.10	77.60	68.41	64.77	76.60	44.31	62.61	61.73	59.41	60.57
MiB [38]	65.51	29.87	61.05	64.53	77.30	**49.57**	63.98	62.40	**60.08**	61.24
ours	**68.16**	78.24	**69.42**	**66.97**	**78.39**	44.34	**64.17**	**65.65**	59.40	**62.53**
Joint	68.26	93.84	71.46	68.76	80.43	67.89	71.46	68.76	74.16	71.46

latest step. However, the performance on the second step of iCaRL is only better than FT, which makes its overall efficacy 10 *p.p* lower than ILT and MiB. Regarding the latest classes, MiB is the best tactic exceeding all data-focused methods by about 5 *p.p*. Our strategy is superior to all others in terms of the old classes and the whole performance, and it achieves comparable results with other methods on the new classes.

It is worth noting that the performance of LWF on the 5s setting on the iSAID dataset reaches 33.89%, only 0.5 *p.p* lower than ours. However, it is 36.37 *p.p* lower on the 4-2s setting of the GCSS dataset. This phenomenon occurs because the performance of LWF heavily depends on the new task data and its degree of relevance to the previous task as analyzed in [18]. Since the background of the iSAID dataset accounts for 97% as shown in Fig. 7, the background still occupies an absolute proportion in the process of CIL. However, the data distribution of the input network changes greatly in the GCSS dataset, which leads to the degradation of LWF performance. The proposed framework performs well on both datasets, which verifies the robustness of our method.

3) Addition of Four Classes in Multistep(4-1s): Table VIII shows the final results on each category under the most challenging 4-1s setting. We can observe that FT and LWF are not enough to prevent CF, and the performance on the old classes (all classes except water) is almost 0%. Though they do quite well with the current category. Combined with Table VII, we can find that the performance of iCaRL in the top four classes has been significantly improved, the mIoU is 62.10% even better than ILT. However, iCaRL has poor performance with mIoU close to 0% in particular categories (road, tree, and vehicle). ILT and MiB have achieved excellent performance in both the old and new classes, and their mIoU both exceeded 60% higher than iCaRL about 20 *p.p*. On all six old classes (since the performance of vehicle class is all 0), our method is far superior to all others. Distinctly, our method

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TABLE VIII
MEAN IOU ON THE GCSS DATASET FOR DIFFERENT CIL METHODS IN THE SETUP OF 4-1S

setting:4-1s	**1-4** (1st step)				**5** (2nd step)	**6** (3rd step)	**7** (4th step)	**8** (5th step)	**mIoU**
Method/class	**building**	**field**	**grass**	**imper. surf.**	**road**	**tree**	**vehicle**	**water**	**all**
FT	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**89.70**	11.21
LWF [35]	0.00	0.00	0.00	0.00	0.00	0.43	0.00	88.30	11.09
iCaRL [19]	80.74	44.88	67.21	55.56	0.00	0.31	0.00	84.91	41.70
ILT [37]	79.18	45.20	66.69	55.85	64.94	85.58	0.00	87.13	60.57
MiB [38]	74.97	47.88	66.34	60.39	66.84	84.74	0.00	88.74	61.24
ours	**81.41**	**49.63**	**69.96**	**61.60**	**68.77**	**86.91**	0.00	81.93	**62.53**
Joint	85.05	51.51	71.53	66.94	72.40	88.47	41.94	93.84	71.46

achieves the best balance between learning new classes and maintaining old knowledge.

4) Analysis of BWT Metric on the GCSS Dataset: Consistent with iSAID, we calculate the BWT scores after the model learned the last task for different increment settings. From Fig. 11, we can discover that FT is still the most severe method for CF. With the increase of the number of tasks for incremental setting, the BWT values of LWF and iCaRL become smaller and smaller, and their memory capacity showed an apparent decreasing trend (the bars are getting longer and longer). ILT, MiB, and our method maintain BWT values no higher than three points in different increment settings, which far surpass the three previously mentioned methods. It can be seen that our method has the best performance in terms of BWT metric under this dataset. Therefore, this again illustrates the importance of using historical information to reconstruct learned classes' labels from background class as well as focusing on the foreground regions of the current learning task to retain old knowledge.

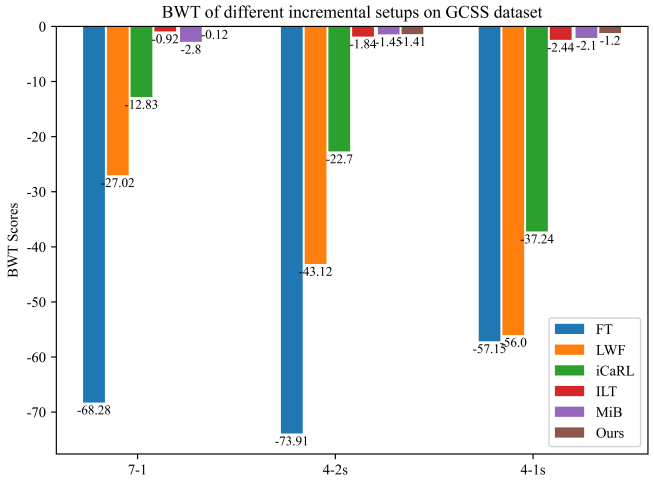


Fig. 11. BWT scores of different methods on the GCSS dataset with different incremental settings.

G. Qualitative Analysis

Because the ratio of background to foreground pixels in the iSAID dataset is 97:3, which results in the visualization effect is not intuitive, therefore, we only show the visualization results on the GCSS dataset. The qualitative results on the 4-1s (5 tasks) of the GCSS dataset are presented in Fig. 12. Except for the new category of water, the remaining seven classes (such as field and building) are all learned previously.

1) Vertical Analysis: From the results in the second and third columns, we can observe that FT and LWF have almost forgotten all the prior knowledge. The bias of their segmentation is completely toward new classes, which is consistent with the results in Table VII. We can discover that iCaRL can effectively reduce CF compared to FT and LWF, but the predictions for some old classes (such as roads and trees) are still inaccurate from the visualizations in the fourth column. The qualitative results in the fifth column manifest that ILT further improves the prediction performance, but the problem of misclassifying the target classes into the background class is still serious. MiB solves the problems of ILT mentioned above by using the background modeling and the classifier initialization strategy, but it introduces the situation that pixels are prone to be misclassified into initialization classes. The results in column six illustrate this phenomenon well, and we can witness that the third and fourth images misdivide road and building into the impervious surfaces, respectively. The visualization results of our method outperform all baselines and are closer to the GT as exhibited in column 7. Compared with ILT, we significantly decrease the number of pixels that misclassified the target categories into the background class. Meanwhile, we reduce the misclassification among target classes and achieve more accurate predictions compared to MiB.

2) Horizontal Analysis: From the qualitative results in the first row, we can observe that the prediction of our method is superior for the learned class of the field (gray). Similarly, our method achieves brilliant results no matter it is the tree class in the second and fifth rows, the road class in the third row, or the buildings and grass classes in the fourth row. Our method can predict the pixels belonging to the old classes superbly after several incremental steps, which fully illustrates the effectiveness of using historical information to reconstruct

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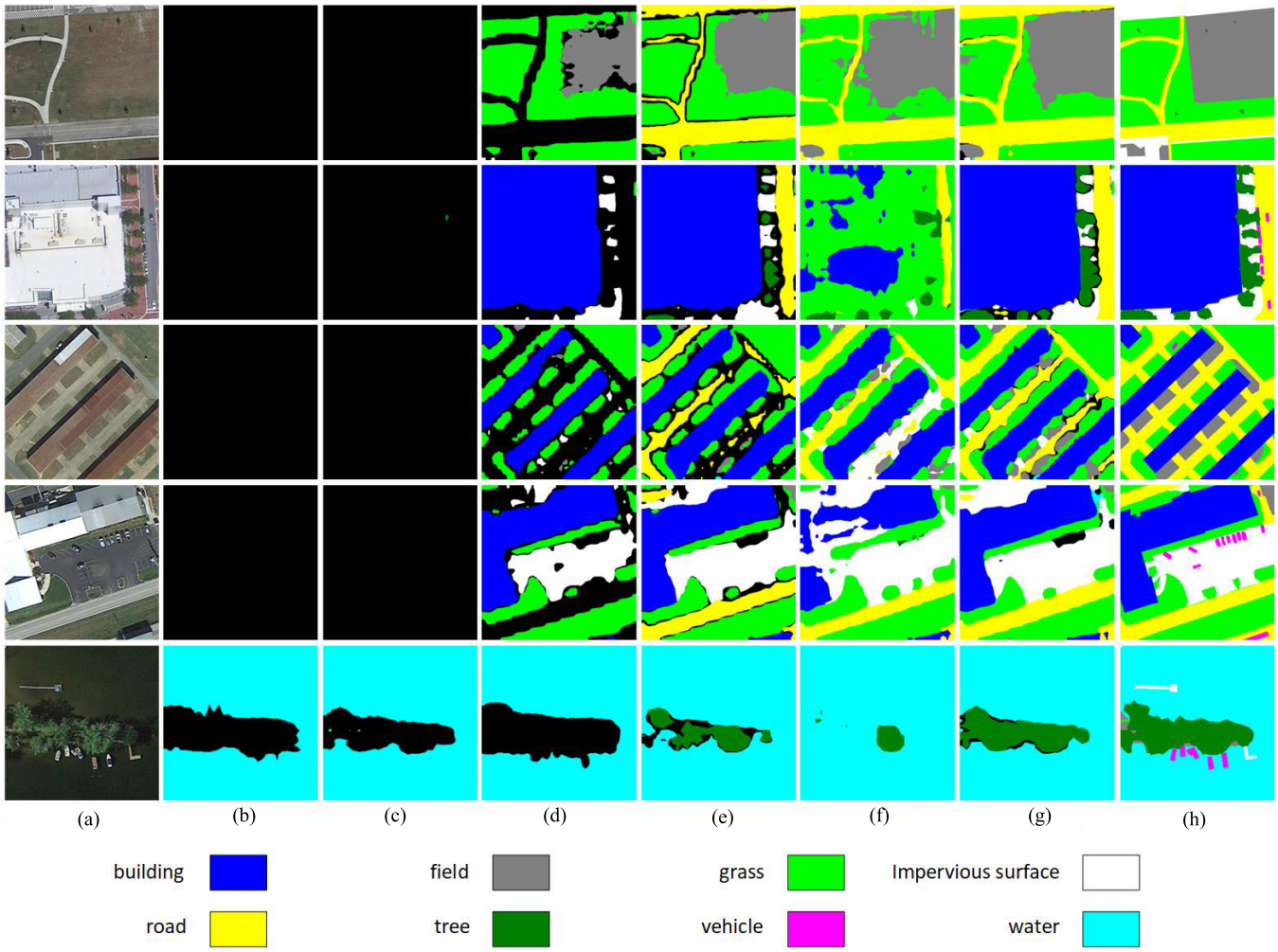


Fig. 12. Visualization of continual semantic segmentation results on the 4-1s (5 steps) setting of the Gaofen dataset. Different columns of pictures from left to right indicate: (a) input image, (b) FT, (c) LWF [35], (d) iCaRL [19], (e) ILT [37], (f) MiB [38], (g) Ours, and (h) GT. The categories represented by the colors in the figure are shown in the legend, where black represents the background class. The qualitative results demonstrate the superiority of our method, especially the old classes (e.g., field, tree, and road).

labels and paying more attention to the foreground region to retain previous knowledge.

H. Failure Cases of Our Approach

Table VIII and Fig. 12 show the failure case of our method. Like other continual learning methods, our approach is not effective for the continuous segmentation of small objects. From Fig. 12(g) and (h), we can see that our method performs very poorly in evaluating the vehicle category. In future work, it is essential to improve the CSS ability of the model for small objects because the small objects are a significant target component of RS images and are difficult to learn. If we can promote the processing power of small objects, it will contribute to the development of CSS in the field of RS.

V. Conclusion

This article proposes a historical information-guided framework to cope with the CF problem in CISS for RS images. In this work, we analyze that historical information is especially important because it is the only source of information related to the old classes when only the new classes are available, while the previous training set is not retained. To overcome the background shift of CISS, we combine historical information with the background class of the current step to retrieve the labels belonging to the old classes. In addition, we design a feature global perception module to strengthen the focus on the foreground region of the input data. This helps to cope with the exacerbated phenomenon of CF caused by complex scenes and imbalance between foreground and background on RS images. Extensive experiments on two datasets that contain very different proportions of foreground and background demonstrate the generality and effectiveness of the proposed framework. It is worth noting that we are the first to establish comparisons between the proposed method and several recently continual learning approaches in CISS on these two datasets. We hope that abundant comparisons with previous continual learning methods on RS images will encourage more work on this burgeoning research topic in the future. As for future work, we will try to promote our method so that it can maintain comparable performance in longer task sequences or can be applied to other dense prediction tasks.

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Meanwhile, we will consider improving our method to work in a more challenging and promising dynamic scene where domains change.

REFERENCES

[1] Y. Ma *et al.*, "Remote sensing big data computing: Challenges and opportunities," *Future Gener. Comput. Syst.*, vol. 51, pp. 47–60, Oct. 2015.

[2] M. Chi, A. Plaza, J. A. Benediktsson, Z. Sun, J. Shen, and Y. Zhu, "Big data for remote sensing: Challenges and opportunities," *Proc. IEEE*, vol. 104, no. 11, pp. 2207–2219, Nov. 2016.

[3] K. Nogueira, M. D. Mura, J. Chanussot, W. R. Schwartz, and J. A. D. Santos, "Dynamic multicontext segmentation of remote sensing images based on convolutional networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7503–7520, Oct. 2019.

[4] L. Ding, J. Zhang, and L. Bruzzone, "Semantic segmentation of large-size VHR remote sensing images using a two-stage multiscale training architecture," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5367–5376, Aug. 2020.

[5] C. Peng, K. Zhang, Y. Ma, and J. Ma, "Cross fusion net: A fast semantic segmentation network for small-scale semantic information capturing in aerial scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.

[6] J. Zhang, S. Lin, L. Ding, and L. Bruzzone, "Multi-scale context aggregation for semantic segmentation of remote sensing images," *Remote Sens.*, vol. 12, no. 4, p. 701, Feb. 2020.

[7] G. Deng, Z. Wu, C. Wang, M. Xu, and Y. Zhong, "CCANet: Class-constraint coarse-to-fine attentional deep network for subdecimeter aerial image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–20, 2022.

[8] D. Zhao, C. Wang, Y. Gao, Z. Shi, and F. Xie, "Semantic segmentation of remote sensing image based on regional self-attention mechanism," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[9] R. Niu, X. Sun, Y. Tian, W. Diao, K. Chen, and K. Fu, "Hybrid multiple attention network for semantic segmentation in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022.

[10] X. Sun, A. Shi, H. Huang, and H. Mayer, "BAS^{4}Net: Boundary-aware semi-supervised semantic segmentation network for very high resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5398–5413, 2020.

[11] L. Ding, H. Tang, and L. Bruzzone, "LANet: Local attention embedding to improve the semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 426–435, Jan. 2021.

[12] L. Ding *et al.*, "Looking outside the window: Wide-context transformer for the semantic segmentation of high-resolution remote sensing images," 2021, *arXiv:2106.15754*.

[13] L. Wang, R. Li, C. Duan, C. Zhang, X. Meng, and S. Fang, "A novel transformer based semantic segmentation scheme for fine-resolution remote sensing images," 2021, *arXiv:2104.12137*.

[14] M. McCloskey and N. J. Cohen, "Catastrophic interference in connectionist networks: The sequential learning problem," *Psychol. Learn. Motivat.*, vol. 24, pp. 109–165, Dec. 1989.

[15] G. M. van de Ven and A. S. Tolias, "Three scenarios for continual learning," 2019, *arXiv:1904.07734*.

[16] A. Chaudhry, P. K. Dokania, T. Ajanthan, and P. H. Torr, "Riemannian walk for incremental learning: Understanding forgetting and intransigence," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 532–547.

[17] R. M. French, "Catastrophic forgetting in connectionist networks," *Trends Cogn. Sci.*, vol. 3, no. 4, pp. 128–135, Apr. 1999.

[18] M. D. Lange *et al.*, "A continual learning survey: Defying forgetting in classification tasks," 2019, *arXiv:1909.08383*.

[19] S.-A. Rebuffi, A. Kolesnikov, G. Sperl, and C. H. Lampert, "ICaRL: Incremental classifier and representation learning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2001–2010.

[20] D. Lopez-Paz and M. Ranzato, "Gradient episodic memory for continual learning," 2017, *arXiv:1706.08840*.

[21] A. Chaudhry, M. Ranzato, M. Rohrbach, and M. Elhoseiny, "Efficient lifelong learning with A-GEM," 2018, *arXiv:1812.00420*.

[22] J. Garcia, "Learning without memory," *J. Cognit. Neurosci.*, vol. 2, no. 4, pp. 287–305, 1990.

[23] A. Douillard, M. Cord, C. Ollion, T. Robert, and E. Valle, "PodNet: Pooled outputs distillation for small-tasks incremental learning," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 86–102.

[24] H. Shin, J. K. Lee, J. Kim, and J. Kim, "Continual learning with deep generative replay," 2017, *arXiv:1705.08690*.

[25] Y. Cong, M. Zhao, J. Li, S. Wang, and L. Carin, "GAN memory with no forgetting," 2020, *arXiv:2006.07543*.

[26] M. Zhai, L. Chen, and G. Mori, "Hyper-LifelongGAN: Scalable lifelong learning for image conditioned generation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2246–2255.

[27] A. A. Rusu *et al.*, "Progressive neural networks," 2016, *arXiv:1606.04671*.

[28] C. Fernando *et al.*, "PathNet: Evolution channels gradient descent in super neural networks," 2017, *arXiv:1701.08734*.

[29] A. Mallya and S. Lazebnik, "PackNet: Adding multiple tasks to a single network by iterative pruning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7765–7773.

[30] A. Mallya, D. Davis, and S. Lazebnik, "Piggyback: Adapting a single network to multiple tasks by learning to mask weights," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 67–82.

[31] X. Li, Y. Zhou, T. Wu, R. Socher, and C. Xiong, "Learn to grow:A continual structure learning framework for overcoming catastrophic forgetting," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 3925–3934.

[32] J. Rajasegaran, M. Hayat, S. Khan, F. S. Khan, and L. Shao, "Random Path Selection for Incremental Learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019.

[33] X. Lu, X. Sun, W. Diao, Y. Feng, P. Wang, and K. Fu, "LIL: Lightweight incremental learning approach through feature transfer for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–20, 2022.

[34] K. James *et al.*, "Overcoming catastrophic forgetting in neural networks," *Proc. Nat. Acad. Sci. USA*, vol. 114, no. 13, pp. 3521–3526, Mar. 2017.

[35] Z. Li and D. Hoiem, "Learning without forgetting," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 12, pp. 2935–2947, Dec. 2018.

[36] S.-W. Lee, J.-H. Kim, J. Jun, J.-W. Ha, and B.-T. Zhang, "Overcoming catastrophic forgetting by incremental moment matching," 2017, *arXiv:1703.08475*.

[37] U. Michieli and P. Zanuttigh, "Incremental learning techniques for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 3205–3212.

[38] F. Cermelli, M. Mancini, S. R. Bulo, E. Ricci, and B. Caputo, "Modeling the background for incremental learning in semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9233–9242.

[39] A. Douillard, Y. Chen, A. Dapogny, and M. Cord, "PLOP: Learning without forgetting for continual semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 4040–4050.

[40] U. Michieli and P. Zanuttigh, "Continual semantic segmentation via repulsion-attraction of sparse and disentangled latent representations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 1114–1124.

[41] A. Maracani, U. Michieli, M. Toldo, and P. Zanuttigh, "RECALL: Replay-based continual learning in semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 7026–7035.

[42] F. Zenke, B. Poole, and S. Ganguli, "Continual learning through synaptic intelligence," in *Proc. Int. Conf. Mach. Learn.*, 2017, pp. 3987–3995.

[43] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," *Comput. Sci.*, vol. 14, no. 7, pp. 38–39, 2015.

[44] S. Waqas Zamir *et al.*, "iSAID: A large-scale dataset for instance segmentation in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, Jun. 2019, pp. 28–37.

[45] X. Sun *et al.*, "Automated high-resolution Earth observation image interpretation: Outcome of the 2020 gaofen challenge," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 8922–8940, 2021.

[46] (2020). *2020 Gaofen Challenge*. Website. [Online]. Available: http://gaofen-challenge.com/

[47] E. Shelhamer, J. Long, and T. Darrell, "Fully convolutional networks for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 640–651, Apr. 2017.

[48] A. Vaswani *et al.*, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 5998–6008.

[49] S. Zheng *et al.*, "Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 6881–6890.

[50] Z. Liu *et al.*, "Swin transformer: Hierarchical vision transformer using shifted Windows," 2021, *arXiv:2103.14030*.

[51] J. Schwarz *et al.*, "Progress & compress: A scalable framework for continual learning," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 4528–4537.

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[52] J. Smith, Y.-C. Hsu, J. Balloch, Y. Shen, H. Jin, and Z. Kira, "Always be dreaming: A new approach for data-free class-incremental learning," 2021, *arXiv:2106.09701*.
[53] Y. Feng, X. Sun, W. Diao, J. Li, X. Gao, and K. Fu, "Continual learning with structured inheritance for semantic segmentation in aerial imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022.
[54] O. Tasar, Y. Tarabalka, and P. Alliez, "Incremental learning for semantic segmentation of large-scale remote sensing data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 9, pp. 3524–3537, Sep. 2019.
[55] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.
[56] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[57] T. Wang, L. Yuan, X. Zhang, and J. Feng, "Distilling object detectors with fine-grained feature imitation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 4933–4942.
[58] J. Guo *et al.*, "Distilling object detectors via decoupled features," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2154–2164.
[59] Y. Feng, X. Sun, W. Diao, J. Li, and X. Gao, "Double similarity distillation for semantic image segmentation," *IEEE Trans. Image Process.*, vol. 30, pp. 5363–5376, 2021.
[60] S. J. Pan, I. W. Tsang, J. T. Kwok, and Q. Yang, "Domain adaptation via transfer component analysis," *IEEE Trans. Neural Netw.*, vol. 22, no. 2, pp. 199–210, Feb. 2010.
[61] Y. Li, T. Shi, Y. Zhang, W. Chen, Z. Wang, and H. Li, "Learning deep semantic segmentation network under multiple weakly-supervised constraints for cross-domain remote sensing image semantic segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 175, pp. 20–33, May 2021.
[62] G. Csurka, D. Larlus, and F. Perronnin, "What is a good evaluation measure for semantic segmentation?" in *Proc. Brit. Mach. Vis. Conf.*, 2013, p. 5244.
[63] S. R. Bulo, L. Porzi, and P. Kontschieder, "In-place activated BatchNorm for memory-optimized training of DNNs," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 5639–5647.



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